

YASHKUL', Russian FederationWMO: **348660**Lat: **46.183N**Lon: **45.350E**Elev: **-7**StdP: **101.41**Time Zone: **3.00 (E03)**Period: **94-15**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.6	-16.7	-23.5	0.5	-20.4	-19.4	0.7	-15.9	11.7	0.1	9.7	-2.3	2.2	0	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	37.9	21.2	36.1	20.8	34.4	20.3	23.3	33.5	22.3	32.3	21.4	31.2	3.9	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	14.9	27.3	19.1	13.9	26.4	18.2	13.1	25.4	69.1	33.5	65.4	32.3	62.2	31.3	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	10.0	8.4	7.2	DB	-23.2	40.3	6.0	2.4	-27.5	42.0	-31.0	43.4	-34.3	44.8	-38.6	46.5
(5)				WB	-23.4	24.4	5.9	1.6	-27.6	25.6	-31.1	26.6	-34.4	27.5	-38.6	28.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	-3.4	-2.8	3.2	11.3	18.5	24.0	26.4	25.3	18.3	11.1	3.5	-0.8
	DBStd	11.67	6.43	6.53	4.26	4.22	4.36	3.32	2.93	3.71	4.15	4.72	5.22	5.37
	HDD10.0	1594	414	358	211	32	2	0	0	1	42	199	334	
	HDD18.3	3313	672	592	468	212	53	2	0	1	49	227	444	592
	CDD10.0	2074	0	0	2	72	265	419	509	474	251	77	4	0
	CDD18.3	752	0	0	0	2	58	172	251	217	49	3	0	0
	CDH23.3	9276	0	0	0	39	686	2038	3166	2774	531	43	0	0
	CDH26.7	4654	0	0	0	4	257	996	1723	1498	169	7	0	0
Wind	WSAvg	2.8	2.7	3.1	3.4	3.1	2.9	2.8	2.6	2.6	2.7	2.6	2.6	2.8
Precipitation	PrecAvg	292	16	13	21	26	39	39	26	21	26	26	20	19
	PrecMax	394	30	22	53	95	92	148	121	81	98	95	49	40
	PrecMin	191	1	2	6	1	1	2	0	1	1	0	5	1
	PrecStd	60	7	6	13	22	28	30	25	19	26	25	13	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	10.5	17.2	27.2	34.9	38.3	40.3	40.1	33.9	27.5	16.8	10.7
		MCWB	6.5	6.9	11.5	16.4	20.2	21.1	20.8	21.7	20.9	16.7	12.5	8.7
	2%	DB	6.7	7.5	14.4	24.3	31.8	36.1	37.9	37.9	30.6	24.2	14.4	8.6
		MCWB	4.9	4.8	8.9	14.9	17.2	20.9	21.0	21.7	19.1	15.9	11.4	7.1
	5%	DB	5.0	5.6	12.2	21.8	29.7	34.3	36.1	35.9	28.6	21.1	12.1	6.9
		MCWB	3.7	3.7	7.6	14.1	17.1	20.1	20.6	21.0	18.1	14.5	9.8	5.5
	10%	DB	3.5	4.2	9.9	19.5	27.5	32.2	34.3	34.0	26.5	18.4	10.4	5.3
		MCWB	2.5	2.8	6.3	12.9	16.9	19.6	20.1	20.4	17.3	13.5	8.7	4.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	7.3	11.8	18.1	22.4	24.1	23.9	25.1	22.7	18.1	13.3	9.2
		MCDB	8.9	9.8	16.4	24.3	33.2	34.1	35.5	35.8	29.9	24.1	15.9	10.3
	2%	WB	5.1	5.1	9.4	16.1	19.3	22.7	22.7	23.6	20.8	16.5	11.7	7.3
		MCDB	6.4	7.0	13.2	22.4	27.9	32.2	34.4	33.5	28.0	21.6	13.8	8.4
	5%	WB	3.8	3.9	8.0	14.6	18.3	21.5	21.7	22.4	19.5	15.3	10.2	5.8
		MCDB	4.8	5.3	11.5	20.7	26.6	30.8	32.6	31.9	26.2	20.2	11.9	6.7
	10%	WB	2.5	2.9	6.6	13.4	17.5	20.4	21.0	21.4	18.2	14.0	8.8	4.4
		MCDB	3.4	4.0	9.7	18.7	25.1	29.4	31.0	31.2	24.5	17.9	10.3	5.3
Mean Daily Temperature Range	5% DB	MDBR	6.0	6.9	9.1	11.4	12.3	12.8	12.9	13.6	12.7	10.1	7.2	5.4
		MCDBR	6.5	7.9	12.8	14.8	16.2	16.2	15.7	16.6	15.3	14.0	9.4	6.7
		MCWBR	4.8	5.8	7.9	7.0	5.7	5.4	4.8	6.1	6.3	6.6	6.5	5.4
	5% WB	MCDBR	6.0	7.2	11.9	13.5	13.9	14.5	13.8	14.7	13.8	12.2	8.6	6.6
		MCWBR	4.7	5.4	7.7	6.8	5.9	6.0	5.3	6.5	6.4	6.5	6.2	5.3
Clear-Sky Solar Irradiance	taub	0.356	0.388	0.419	0.475	0.447	0.439	0.451	0.457	0.433	0.393	0.366	0.365	
	taud	2.167	2.098	2.085	1.975	2.137	2.214	2.190	2.151	2.162	2.235	2.224	2.154	
	Ebn at Noon	700	754	793	782	824	831	815	791	771	745	685	636	
	Edh at Noon	91	118	138	169	149	139	140	140	127	102	85	83	
All-Sky Solar Radiation	RadAvg	1.03	1.86	3.13	4.79	6.03	6.66	6.50	5.79	4.30	2.62	1.28	0.80	
	RadStd	0.23	0.33	0.36	0.41	0.47	0.37	0.35	0.33	0.32	0.32	0.20	0.10	